

IAPR 2025 Final Project

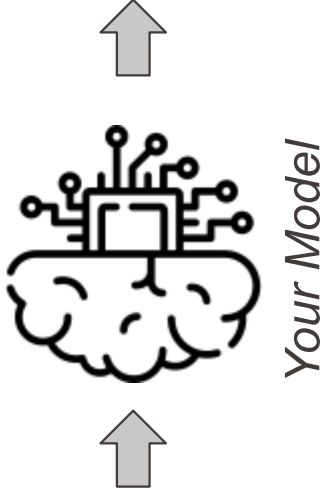
Chocolate Recognition

Introduction

- Implement a chocolate detector that can count chocolates of predefined types in an image



Input Image



Category	Count
Jelly White	1
Jelly Milk	0
Jelly Black	0
Amandina	0
Crème brûlée	0
Triangolo	0
Tentation Noir	0
Comtesse	1
Noblesse	0
Noir authentique	0
Passion au lait	1
Arabia	1
Stracciatella	1

Prediction

- 13 reference images
- 1 per chocolate category



Jelly White



Jelly Milk



Jelly Black



Amandina



Crème brûlée



Tentation
Noir



Arabia



Comtesse



Noblesse



Triangolo



Passion au lait



Stracciatella



Noir authentique

- 270 images
 - 90 training images with weak labels
 - 180 testing images



All photos were taken under the same conditions — using the same camera, at the same distance, and with the same resolution.

- 7 different backgrounds
 - Evenly distributed in train/test



White



Sweets Box (verso)



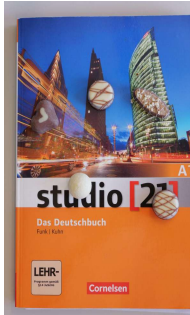
Starter Kit



Bag



Plastic



Book



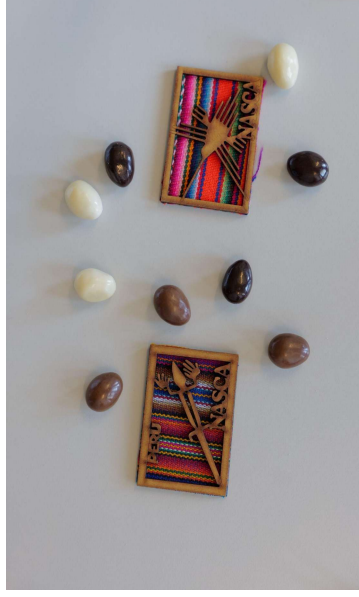
Sweets Box (recto)

Dataset

- Half of images contain other objects
- Evenly distributed across splits & backgrounds



Without additional objects



With additional objects



Test set may contain objects not present in the train set.

- In Kaggle, you will
 - Access training and test data
 - Submit test predictions and monitor your progress on the **public leaderboard**
 -  **Maximum 5 submission per team per day**
 - Compete for fun (and chocolate!)
- **Two Challenges**
 - **Classic**—no deep learning
 - **Deep Learning**—limited in resources



CAREFULLY READ THE RULES OF EACH CHALLENGE

You must choose **one** challenge to participate in

You are free to switch challenge during the contest

- Each challenge has **two leaderboards**
 - **Public Leaderboard**
 - Visible during the competition
 - Based on 50% of the test set
 - **Private Leaderboard**
 - Revealed after the competition ends
 - Based on the remaining 50%
 **Counts for grading**



Performance Goal:

- Beat the TA's baseline on the **private leaderboard**

PublicPrivate

The private leaderboard is calculated with approximately 69% of the test data. This competition has completed. This leaderboard reflects the final standings.

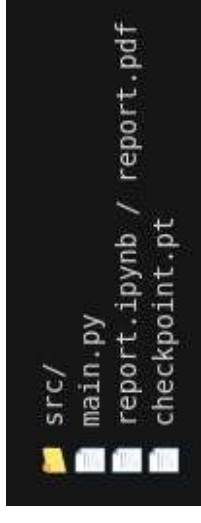
#	Δ	Team	Members
1	—	Group 21	  
2	+ 3	group17	  
3	+ 3	ba3ba3	  

Ranking of last year

Baseline

Submission Instructions

- You must submit by **May 21st (11:59pm)** a .zip with:
 - **Jupyter notebook report** (report.ipynb/pdf)
 - Clear justification of design choices
 - Detailed technical descriptions of your solution
 - Strong quantitative and qualitative analysis, demonstrating and explaining the ability of your solution to segment the chocolates from the background and to compute relevant object descriptors
- **Executable (main.py)**
 - Produce the exact final submission file you uploaded to Kaggle
 - Use only Python packages covered in Labs 1–3 (Python 3.9).
- ⚠ **The reproducibility of your Kaggle submission is part of the grade!**
- **Source code (src/)**
 - Include all additional code and files used by main.py.
 - If using deep learning, you may also include your model checkpoints, ensuring they can be loaded correctly.



Final Presentation

- Format
 - **7-minutes** presentation
 - Summarize your approach and design decisions
 - Highlight your main results (quantitative+qualitative)
 - **8-minutes** Q&A with the TAs
 - At least one question per person
 - TAs will ask targeted questions



Date: Friday, May 23rd
Time: Between 8:00 AM – 1:00 PM
Location: In person

Time	ELE 242	ELD 138	ELD 120	DIA 003
8h00	Group 22	Group 50	Group 44	Group 43
8h15	Group 14	Group 10	Group 9	Group 2
8h30	Group 37	Group 4	Group 52	Group 55
8h45	Group 75	Group 64	Group 46	Group 47
9h15	Group 23	Group 39	Group 1	Group 12
9h30	Group 5	Group 40	Group 34	Group 35
9h45	Group 33	Group 80	Group 74	Group 31
10h00	Group 13	Group 24	Group 26	Group 77
10h30	Group 7	Group 32	Group 38	Group 19
10h45	Group 69	Group 20	Group 48	Group 51
11h00	Group 15	Group 8	Group 73	Group 17
11h15	Group 18	Group 67	Group 25	Group 66
11h45	Group 21	Group 41	Group 56	Group 16
12h00	Group 11	Group 3	Group 36	Group 27
12h15	Group 49	Group 42	Group 29	Group 30
12h30	Group 60	Group 28		

Schedule

- 100 points

Category	Criteria	Points
1. Report	Design justification	20
	Technical implementation	10
	Qualitative analysis (segmentation/object description)	30
2. Kaggle Performance	Beat TA's baseline in your challenge	10
3. Reproducibility	Executable produces same submission as on Kaggle	10
4. Presentation	Clarity of presentation and quality of answers	20
5. Bonus / Penalty	Top-5 ranking bonus / rule violation penalty	±X

Bonus Points

- Attributed for each challenge

Rank	Bonus Points
 1st	+10 + 
 2nd	+7
 3rd	+5
4th	+3
5th	+1

Questions ?